

Survival MF Derivations - Yusha

Andrew Walther

January 21, 2026

1 Introduction

The following document presents a derivation of the objective function & algorithmic updates for supervised survival matrix factorization by Yusha Liu.

2 Objective Function

$$F(q_l, q_f, q_\beta, g_l, g_f, \beta, \tau) = E_{q_l, q_f, q_\beta} [\log p(Y, t, \delta | L, F, \beta, \tau)] + \bar{z}_k E_{q_{l_k}} \left[\log \frac{g_{l_k}(l_k)}{q_{l_k}(l_k)} \right] + \bar{z}_k E_{q_{f_k}} \left[\log \frac{g_{f_k}(f_k)}{q_{f_k}(f_k)} \right] + \bar{z}_k E_{q_{\beta_k}} \left[\log \frac{g_{\beta_k}(\beta_k)}{q_{\beta_k}(\beta_k)} \right] \quad (1)$$

$$\log P(Y, t, \delta | L, F, \beta, \tau) = \log P(Y | L, F, \sigma^2) \log P(t, \delta, L, \beta) \quad (2)$$

$$l(\eta) = \sum_i \delta_i \left[\eta_i - \log \left(\sum_{j \in R(t_i)} \exp(\eta_j) \right) \right] \quad (3)$$

$$= \underbrace{(\hat{\eta}^{(b-1)})}_{\text{pink}} + \underbrace{(\eta - \hat{\eta}^{(b-1)}) \nabla l(\hat{\eta}^{(b-1)})}_{\text{pink}} + \underbrace{\frac{1}{2} (\eta - \hat{\eta}^{(b-1)})' \nabla^2 (\hat{\eta}^{(b-1)}) (\eta - \hat{\eta}^{(b-1)})}_{\text{pink}} \quad (4)$$

$$= \underbrace{(\hat{\eta}^{(b-1)})}_{\text{pink}} + \underbrace{\sum_i u_i^{(b-1)} (\eta_i - \hat{\eta}_i^{(b-1)})}_{\text{pink}} + \underbrace{\frac{1}{2} \sum_i H_{ii}^{(b-1)} (\eta_i - \hat{\eta}_i^{(b-1)})^2}_{\text{pink}} \quad (5)$$

$$= - \sum_i \frac{1}{2} W_{ii} \left[\eta_i - \left(\hat{\eta}_i + \frac{u_i}{W_{ii}} \right)^2 \right] + u_i (\eta_i - \hat{\eta}_i) - \frac{1}{2} W_{ii} (\eta_i - \hat{\eta}_i)^2 \text{ (allowing } W_{ii} = -H_{ii}) \quad (6)$$

$$= -\frac{1}{2} W_{ii} [\eta_i^2 - 2\eta_i \hat{\eta}_i + \hat{\eta}_i^2 - 2W_{ii}^{-1} u_i (\eta_i - \hat{\eta}_i)] \quad (7)$$

$$= -\frac{1}{2} W_{ii} \left[\eta_i^2 - 2 \left(\hat{\eta}_i + \frac{u_i}{W_{ii}} \eta_i + \dots \right) \right] \quad (8)$$

$$= -\frac{1}{2} W_{ii} \left[\eta_i - \left(\hat{\eta}_i + \frac{u_i}{W_{ii}} \right) \right]^2 + \dots \quad (9)$$

$$\mathcal{F}(q_l, q_f, \underline{q_\beta}, g_l, g_f, \underline{\beta}, \tau) = E_{q_l, q_f, q_\beta} [\log P(Y | L, F, \tau) \log P(t, \delta | L, \beta)] + \sum_k E_{q_{l_k}} [\dots] + \sum_k E_{q_{f_k}} [\dots] + \dots \quad (10)$$

relative to EBMF A.3-A.4, include qB and B

$$= E_{q_l, q_f, q_\beta} \left[\bar{z}_i \bar{z}_j - \frac{1}{2} \tau_j (y_{ij} - \bar{z}_k l_{ik} f_{jk})^2 + \bar{z}_i - \frac{1}{2} W_{ii}^{(b-1)} \left(z_i^{(b-1)} - \sum_k l_{ik} \beta_k \right)^2 \right] + \dots \quad (11)$$

2A Update q_{l_k} (A.35 - A.40)

$$F(q_l, q_f, q_\beta, g_l, g_f, g_\beta, \tau) = E_{q_{l_k}} \left[-\frac{1}{2} \bar{z}_i (A_{ik} l_{ik}^2 - 2B_{ik} l_{ik}) \right] + E_{q_{l_k}} \left[\log \frac{g_{l_k}(l_k)}{q_{l_k}(l_k)} \right] + Constant \quad (12)$$

- $A_{ik} = ???$
- $B_{ik} = ???$
- Simplified to EBNM problem
- Drop the superscript $(b-1)$ for convenience.

$$W_{ii} (z_i - \bar{z}_k l_{ik} \beta_k)^2 = W_{ii} (l_{ik}^2 \beta_k^2 - 2\bar{z}_k l_{ik} \beta_k) + Constant \quad (13)$$

where

$$\tilde{z}_{ik} = z_i - \bar{z}_{k' \neq k} E_q [l_{ik'} \beta_{k'}] = z_i - \bar{z}_{k' \neq k} \bar{l}_{ik'} \bar{\beta}_{k'} \quad (14)$$

2B Update q_{f_k}

$$F(q_l, q_f, q_\beta, g_l, g_f, g_\beta, \tau) = E_{q_{f_k}} \left[-\frac{1}{2} \bar{z}_j (\tilde{A}_{jk} f_{jk}^2 - 2\tilde{\beta}_{jk} f_{jk}) \right] + E_{q_{f_k}} \left[\log \frac{g_{f_k}(f_k)}{q_{f_k}(f_k)} \right] + Constant \quad (15)$$

- $\tilde{A}_{jk} = ???$
- $\tilde{\beta}_{jk} = ???$

2C Update q_β

Let

- $\bar{l}_{ik} = E_{q_l} [l_{ik}]$
- $\bar{l}_{ik}^2 = E_{q_l} [l_{ik}^2]$
- $\bar{l}_{ik}^2 = (\bar{l}_{ik})^2 + Var_{q_l} [l_{ik}]$

$$F(q_l, q_f, q_\beta, g_l, g_f, g_\beta, \tau) \quad (16)$$

$$= E_{q_l, q_\beta} \left[-\frac{1}{2} \bar{z}_i W_{ii}^{(b-1)} \left(\bar{z}_k l_{ik}^2 \beta_k^2 + 2\bar{z}_{k \neq k'} l_{ik} \beta_k l_{ik'} \beta_{k'} - 2z_i^{(b-1)} \bar{z}_k l_{ik} \beta_k \right) \right] \quad (17)$$

$$+ E_{q_\beta} \left[\log \frac{g_\beta(\beta)}{q_\beta(\beta)} \right] + Constant \quad (18)$$

$$= E_{q_\beta} \left[-\frac{1}{2} \sum_i W_{ii}^{(b-1)} \left(\bar{z}_k \bar{l}_{ik}^2 \beta_k^2 + 2\bar{z}_{k \neq k'} \bar{l}_{ik} \bar{l}_{ik'} \beta_k \beta_{k'} - 2z_i^{(b-1)} \bar{z}_k \bar{l}_{ik} \beta_k \right) \right] \quad (19)$$

$$+ E_{q_\beta} \left[\log \frac{g_{beta}(\beta)}{q_{beta}(\beta)} \right] + Constant \quad (20)$$

$$= E_{q_\beta} \left[-\frac{1}{2} \bar{z}_i W_{ii}^{(b-1)} \left(\bar{z}_k (\bar{l}_{ik})^2 \beta_k^2 + 2\bar{z}_{k \neq k'} \bar{l}_{ik} \bar{l}_{ik'} \beta_k \beta_{k'} - 2z_i^{(b-1)} \sum_k \bar{l}_{ik} \beta_k \right) - \frac{1}{2} \bar{z}_i W_{ii}^{(b-1)} (\bar{z}_k Var_{q_l} (l_{ik}) \beta_k^2) \right] \quad (21)$$

$$+ E_{q_\beta} \left[\log \left(\pi_k \frac{1}{\sqrt{2\pi\sigma}} \exp \left(-\frac{1}{2\sigma^2} \beta_k^2 \right) \right) \right] - E_{q_\beta} [\log q(\beta)] \quad (22)$$

$$= E_{q_\beta} \left[-\frac{1}{2} \bar{z}_i W_{ii}^{(b-1)} \left(z_i^{(b-1)} - \bar{z}_k \bar{l}_{ik} \beta_k \right)^2 \right] \quad (23)$$

$$+ E_{q_\beta} \left[-\frac{1}{2} \bar{z}_k \left(\frac{1}{\sigma^2} + \bar{z}_i W_{ii}^{(b-1)} Var_{q_l} (l_{ik}) \right) \beta_k^2 \right] - E_{q_\beta} [\log q(\beta)] \quad (24)$$

This is the same objective function as

$$z_i^{(b-1)} = \bar{z}_k \bar{l}_{ik} \beta_k + \varepsilon_i, \quad \varepsilon_i \sim N \left(0, \frac{1}{W_{ii}^{(b-1)}} \right) \quad (25)$$

with the prior

$$\beta_k \sim N \left(0, \frac{1}{\frac{1}{\sigma^2} + \bar{z}_k W_{ii}^{(b-1)} \text{Var}_{q_l}(l_{ik})} \right) \quad (26)$$

Recall that $\eta_i = \bar{z}_k l_{ik} \beta_k$ so after iteration (b), update $\hat{\eta}_i^{(b)} = E_{q_l, q_\beta} [\bar{z}_k l_{ik} \beta_k] = \bar{z}_k \bar{l}_{ik} \bar{\beta}_k$

2D Regression Model

Consider the linear regression model

$$z_i = X_i' \beta + \varepsilon_i, \quad \varepsilon_i \sim N(0, W_{ii}^{-1}) \quad (27)$$

where the prior is

$$\beta_k \stackrel{\text{iid}}{\sim} N(0, \sigma^2) \Rightarrow \beta | \sigma^2 \sim N(0, \sigma^2 I_{k \times k}) \quad (28)$$

and posterior

$$q(\beta) \sim N(m, V). \quad (29)$$

In an EB framework

$$F(q_\beta, g_\beta) \quad (30)$$

$$= E_q [\log P(z|\beta)] + E_q \left[\log \frac{g_\beta(\beta)}{q_\beta(\beta)} \right] \quad (31)$$

$$= E_q \left[\log \left(\pi_i \frac{1}{\sqrt{2\pi}} W_{ii}^{\frac{1}{2}} \exp \left[-\frac{1}{2} W_{ii} (z_i - x_i' \beta)^2 \right] \right) \right] + E_q \left[\log \left(\pi_k \frac{1}{\sqrt{2\pi\sigma}} \exp \left[-\frac{1}{2\sigma^2} \beta_k^2 \right] \right) \right] \quad (32)$$

$$- E_q [\log q(\beta)] \quad (33)$$

$$= E_q \left[-\frac{1}{2} \bar{z}_i W_{ii} (z_i - x_i' \beta)^2 - \frac{1}{2} \frac{1}{\sigma^2} \bar{z}_k \beta_k^2 - \log q_\beta(\beta) \right] \quad (34)$$